DAY 02: MEASURES OF CENTRAL TENDENCY (MEAN)

Bilingual Study Notes (English & Hindi) | Mission 2029

1. What is Mean? (Mean Kya Hai?)

Mean is the central value of a data set, calculated by adding all values and dividing by the total number of values.

Mean kisi bhi data set ki "Beech ki value" (central value) hoti hai, jo sabhi numbers ko jod kar unki kul ginti se bhag (divide) karne par milti hai.

2. The Formula (Sutra)

$$\text{Mean } (\bar{x}) = \Sigma X / n$$

Where ΣX is the sum of all values and n is the total number of observations.

Jahan ΣX sabhi values ka jod hai aur n unki kul ginti hai.

3. Practical Case Study (Asli Udaharan)

Imagine you produced 5 gold rings today with weights: 5g, 6g, 4g, 7g, and 3g.

Maaniye aapne aaj 5 sone ki rings banayi jinka wazan hai: 5g, 6g, 4g, 7g, aur 3g.

Step 1 (Sum): 5 + 6 + 4 + 7 + 3 = 25 grams.

Step 1 (Jod): *5 + 6 + 4 + 7 + 3 = 25 grams.*

Step 2 (Count): Total rings (n) = 5.

Step 2 (Ginti): *Total rings (n) = 5.*

Step 3 (Mean): 25 / 5 = 5 grams.

Step 3 (Mean): *25 / 5 = 5 grams.*

The "Origin" Secret: The Weakness of Mean

The "Origin" Secret: Mean ki Kamzori

Mean is very sensitive to "Outliers" (Extreme values). If one ring was 100g, it would change the average from 5g to 24g, which is misleading.

Mean "Outliers" (bahut badi ya chhoti values) ke prati bahut sensitive hota hai. Agar ek ring 100g ki ho jaye, toh average 5g se badhkar 24g ho jayega, jo ki galat tasveer dikhayega.

4. Why it matters for Analysts? (Analysts ke liye kyun zaroori?)

It helps to find the "Typical" behavior of business data, like average sales or average production time.

Ye business data ka "Aam Bartav" samjhne mein madad karta hai, jaise ausat bikri ya ausat production time.